

B.TECH FIRST YEAR | FIRST SEMESTER | ENGINEERING
MATHEMATICS

Engineering Mathematics

Unit V

Vector Integration

Complete Study Module | Exam-Oriented | Beginner-Friendly

Line Integrals

Surface Integrals

Green's Theorem

Stokes' Theorem

Divergence Theorem

40+ Solved Examples

50 Exercise Problems



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Introduction to Vector Integration

1.1 What is Vector Integration?

In earlier courses, you studied scalar integration — finding the area under a curve, or the total distance traveled. In engineering, however, we often deal with **vector quantities** — quantities that have both a magnitude AND a direction, such as force, velocity, electric field, magnetic field, and fluid velocity.

Vector Integration is the process of integrating vector fields over curves, surfaces, or volumes. It allows us to calculate physically meaningful quantities such as:

- The **work done** by a force along a curved path
- The **flux** (amount of fluid, electric field, etc.) passing through a surface
- The **circulation** of a fluid around a closed curve
- The total **charge** or **mass** within a volume

1.2 Why is Vector Integration Important for Engineers?

Field	Application of Vector Integration
Electrical Engineering	Calculating electric flux using Gauss's Law (Divergence Theorem)
Fluid Mechanics	Finding flow rate across a surface (Surface Integral)
Mechanical Engineering	Computing work done by variable forces along curved paths (Line Integral)
Electromagnetics	Faraday's Law uses Stokes' Theorem to relate EMF to magnetic flux
Civil Engineering	Finding moments and forces on curved dam surfaces

1.3 Overview of Unit V Topics

This unit covers three major types of vector integrals and three major theorems that connect them:

Type of Integral	Integrated Over	Notation	Physical Example
Line Integral	A curve C	$\int_C \mathbf{F} \cdot d\mathbf{r}$	Work done by a force

Surface Integral	A surface S	$\iint_S \mathbf{F} \cdot d\mathbf{S}$	Fluid flux through a surface
Volume Integral	A volume V	$\iiint_V f dV$	Total mass in a region

Theorem	Connects	Physical Meaning
Green's Theorem	Line integral (2D) ↔ Double integral	Relates boundary circulation to area curl
Stokes' Theorem	Line integral (3D) ↔ Surface integral	Circulation = Curl flux through surface
Divergence Theorem	Surface integral ↔ Volume integral	Total outward flux = divergence integrated over volume

1.4 Key Vector Concepts (Quick Recap)

Vector Field $\mathbf{F}(x, y, z)$:

A function that assigns a vector to every point in space. For example, $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ where P, Q, R are scalar functions of x, y, z.

Gradient (∇f):

For a scalar f: $\nabla f = (\partial f/\partial x)\mathbf{i} + (\partial f/\partial y)\mathbf{j} + (\partial f/\partial z)\mathbf{k}$

Divergence ($\nabla \cdot \mathbf{F}$):

$\nabla \cdot \mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z$

Curl ($\nabla \times \mathbf{F}$):

$\nabla \times \mathbf{F} = (\partial R/\partial y - \partial Q/\partial z)\mathbf{i} - (\partial R/\partial x - \partial P/\partial z)\mathbf{j} + (\partial Q/\partial x - \partial P/\partial y)\mathbf{k}$

Line Integrals – Concept and Definition

2.1 What is a Line Integral?

Imagine you are pushing a heavy box along a *curved path* from point A to point B. At every point of the path, the force you apply may be different in magnitude and direction. A regular (scalar) integral cannot handle this — we need a **Line Integral**.

A **Line Integral** integrates a function (scalar or vector) along a curve in space. The curve is called the **path of integration** and it may be in 2D (xy-plane) or 3D (xyz-space).

Formal Definition – Scalar Line Integral:

Let $f(x, y, z)$ be a scalar function defined on a smooth curve C . Divide C into n small segments, each of arc length Δs_k . The scalar line integral is:

$$\int_C f \, ds = \lim_{n \rightarrow \infty} \sum f(x_k, y_k, z_k) \Delta s_k$$

Formal Definition – Vector Line Integral:

Let $\mathbf{F}(x, y, z) = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ be a vector field. The vector line integral along curve C is:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (P \, dx + Q \, dy + R \, dz)$$

where $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$ is the differential displacement vector along C .

2.2 Types of Line Integrals

Type	Formula	What is integrated?
Scalar (arc length)	$\int_C f \, ds$	Scalar function over arc length
With respect to x	$\int_C P \, dx$	x-component along x-displacement
With respect to y	$\int_C Q \, dy$	y-component along y-displacement
With respect to z	$\int_C R \, dz$	z-component along z-displacement
Vector (dot product)	$\int_C \mathbf{F} \cdot d\mathbf{r}$	Total work / circulation

2.3 Properties of Line Integrals

- **Linearity:** $\int_C (a\mathbf{F} + b\mathbf{G}) \cdot d\mathbf{r} = a \int_C \mathbf{F} \cdot d\mathbf{r} + b \int_C \mathbf{G} \cdot d\mathbf{r}$
- **Additivity:** $\int_C = \int_{C_1} + \int_{C_2}$ (when $C = C_1 + C_2$)
- **Reversal of path:** $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$
- **Closed path:** Written as $\oint_C \mathbf{F} \cdot d\mathbf{r}$ (circle on integral)

2.4 Path Independence and Conservative Fields

Conservative Vector Field:

A vector field $\mathbf{F}$ is called conservative if $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints A and B, and NOT on the path taken. This happens when $\mathbf{F} = \nabla\phi$ (gradient of some scalar ϕ), i.e., $\text{curl } \mathbf{F} = \mathbf{0}$.

Note: For a conservative field: $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for any closed curve C.

Work Done and Circulation

3.1 Work Done by a Vector Force Along a Curve

Recall from Physics: When a constant force $\mathbf{F}$ acts on an object moving through displacement $\mathbf{d}$, the work done is $W = \mathbf{F} \cdot \mathbf{d} = |\mathbf{F}||\mathbf{d}|\cos\theta$.

But in real engineering scenarios, the force may vary from point to point, and the path may be curved. In such cases:

Work Done by Force $\mathbf{F}$ along curve C :

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (F_x dx + F_y dy + F_z dz)$$

Here, $d\mathbf{r}$ is the infinitesimal displacement along the curve, and the dot product picks up only the component of force in the direction of motion.

Physically: at each tiny segment of the path, we multiply the component of force along the direction of motion with the length of that segment, then sum everything up. That is the total work done.

3.2 Physical Significance

- If $W > 0$: Force aids the motion (energy is transferred to the object)
- If $W < 0$: Force opposes the motion (energy is removed from the object)
- If $W = 0$: Force is perpendicular to motion at every point (e.g., magnetic force on a charge)

3.3 Circulation

Circulation of a Vector Field:

When the path C is a *closed curve*, the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ is called the **circulation** of $\mathbf{F}$ around C .

$$\text{Circulation} = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

In fluid mechanics, if $\mathbf{F}$ represents fluid velocity, the circulation measures the tendency of the fluid to rotate or swirl around the closed path.

- Positive circulation: fluid rotates counterclockwise (by convention)
- Zero circulation: no net rotation (field may be irrotational)
- Negative circulation: fluid rotates clockwise

3.4 Relationship Between Circulation and Curl

The circulation is directly related to the *curl* of the vector field. Stokes' Theorem (covered in Section 9) formally states this relationship. Intuitively, if the curl of a field is large in some region, the circulation around a loop enclosing that region is also large.

$$\text{Circulation} = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} \quad (\text{Stokes' Theorem})$$

Parametric Representation of Curves

4.1 Why Do We Need Parametrization?

To evaluate a line integral, we must convert it into a regular (single variable) integral. To do this, we express the coordinates x, y, z as functions of a single parameter t . This is called **parametrization**.

Once we parametrize, we replace: x with $x(t)$, y with $y(t)$, z with $z(t)$, and $dx = x'(t)dt$, $dy = y'(t)dt$, $dz = z'(t)dt$. The integral becomes a standard integral in t .

4.2 Common Parametrizations

Curve Type	Parametrization	Parameter Range
Straight line from $A(x_1,y_1)$ to $B(x_2,y_2)$	$x = x_1 + t(x_2 - x_1), y = y_1 + t(y_2 - y_1)$	$t: 0 \text{ to } 1$
Circle $x^2 + y^2 = a^2$ (xy-plane)	$x = a \cos t, y = a \sin t, z = 0$	$t: 0 \text{ to } 2\pi$
Ellipse $x^2/a^2 + y^2/b^2 = 1$	$x = a \cos t, y = b \sin t$	$t: 0 \text{ to } 2\pi$
Parabola $y = x^2$	$x = t, y = t^2$	$t: \text{varies}$
Helix	$x = a \cos t, y = a \sin t, z = bt$	$t: 0 \text{ to } 2\pi n$
Along x-axis ($y=0, z=0$)	$x = t, y = 0, z = 0$	$t: a \text{ to } b$

4.3 Step-by-Step Procedure for Line Integral Evaluation

Step	Action	What to Write
1	Identify curve C and endpoints	State the curve equation, start and end points
2	Choose parameter t	Express x, y, z as functions of t
3	Find derivatives	Compute $dx/dt, dy/dt, dz/dt$
4	Find limits for t	What value of t gives start? What gives end?

5	Substitute into integral	Replace x,y,z and dx,dy,dz with t-expressions
6	Simplify integrand	Combine and simplify the expression in t
7	Integrate	Evaluate the resulting single-variable integral
8	Apply limits	Substitute t limits and compute final answer

4.4 Worked Illustration: Parametrizing a Line Segment

Example: Parametrize the line segment from A(0,0,0) to B(1,2,3).

Solution: The direction vector is $B - A = (1,2,3)$. Using t from 0 to 1:

$$x = 0 + t(1) = t$$

$$y = 0 + t(2) = 2t$$

$$z = 0 + t(3) = 3t$$

So: $dx = dt$, $dy = 2dt$, $dz = 3dt$

When $t=0$: $(x,y,z) = (0,0,0) = A$. When $t=1$: $(x,y,z) = (1,2,3) = B$. ✓

Surface Integrals – Concept and Definition

5.1 What is a Surface Integral?

Just as a line integral integrates over a curve, a **surface integral** integrates over a surface (like the surface of a sphere, a cylinder, or a plane region). Surface integrals arise naturally when calculating the total flux of a field passing through a surface.

Scalar Surface Integral:

For a scalar function $f(x,y,z)$ over surface S :

$$\iint_S f dS$$

This integrates f over the surface with dS = area element of S .

Vector Surface Integral (Flux Integral):

For vector field $\mathbf{F}$ over surface S with unit outward normal $\mathbf{n}$:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS$$

where $d\mathbf{S} = \mathbf{n} dS$ is the vector area element.

5.2 Surface Element dS

For different surfaces, the surface element dS takes different forms:

Surface	dS (vector area element)	Note
$z = g(x,y)$ (explicit)	$(-\partial g/\partial x \mathbf{i} - \partial g/\partial y \mathbf{j} + \mathbf{k}) dx dy$	Normal points upward
$z = \text{const}$ (horizontal plane)	$\mathbf{k} dx dy$	Normal is $(0,0,1)$
$x = g(y,z)$ (yz-plane)	$\mathbf{i} dy dz$	Normal is $(1,0,0)$
$y = g(x,z)$ (xz-plane)	$\mathbf{j} dx dz$	Normal is $(0,1,0)$
Sphere $x^2+y^2+z^2=a^2$	$a^2 \sin\phi d\phi d\theta$ (in spherical coords)	Outward normal = $\mathbf{r}/ \mathbf{r} $

5.3 Unit Normal Vector $\mathbf{n}$

The unit normal vector $\mathbf{n}$ is perpendicular to the surface at every point. For a surface $F(x,y,z) = 0$:

$$\mathbf{n} = \nabla F / |\nabla F|$$

The choice of direction (outward or inward) depends on the problem context. For closed surfaces, "outward" normal is standard.

5.4 Orientation of a Surface

A surface is called **orientable** if we can consistently assign a normal direction to every point. Most surfaces in engineering (planes, spheres, cylinders) are orientable. For closed surfaces, the outward normal convention is used.

Flux Through a Surface

6.1 Physical Meaning of Flux

Imagine water flowing through a net. The **flux** measures how much water passes through the net per unit time. If the net is tilted, less water passes through it (it depends on the angle between the flow direction and the net's normal).

Flux of Vector Field $\mathbf{F}$ through surface S :

$$\Phi = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

This measures the net amount of $\mathbf{F}$ passing through S in the direction of $\mathbf{n}$.

6.2 Computing Flux: Standard Method

For a surface $z = g(x,y)$ with domain D in the xy -plane:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D [\mathbf{F} \cdot (-g_x \mathbf{i} - g_y \mathbf{j} + \mathbf{k})] \, dx \, dy$$

where $g_x = \partial g / \partial x$ and $g_y = \partial g / \partial y$.

6.3 Flux Sign Conventions

- **Positive flux:** Field flows in the same general direction as the outward normal (field exits the surface)
- **Negative flux:** Field flows opposite to the outward normal (field enters the surface)
- **Zero flux:** Field is tangential to the surface everywhere (no net flow through surface)

6.4 Electric Flux (Gauss's Law Preview)

In electrostatics, the electric flux through a closed surface S is:

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q_{\text{enclosed}} / \epsilon_0$$

This is Gauss's Law, which follows directly from the Divergence Theorem (see Section 12).

Green's Theorem (Without Proof)

7.1 Statement of Green's Theorem

Green's Theorem

Let C be a simple, closed, piecewise-smooth curve in the xy -plane, traversed counterclockwise (positively). Let D be the region enclosed by C . If $P(x,y)$ and $Q(x,y)$ have continuous first partial derivatives on an open region containing D , then:

$$\oint_C (P \, dx + Q \, dy) = \iint_D (\partial Q / \partial x - \partial P / \partial y) \, dA$$

Left side: Line integral around the boundary

Right side: Double integral over the enclosed region

7.2 What Green's Theorem Tells Us

Green's theorem establishes a beautiful relationship: the circulation of a vector field around a closed boundary equals the integral of the "curl" (specifically, $\partial Q / \partial x - \partial P / \partial y$) over the enclosed area.

It converts a hard line integral into (sometimes) an easier double integral, or vice versa.

7.3 Conditions for Green's Theorem

- Curve C must be **closed** (start and end at same point)
- Curve C must be **simple** (does not cross itself)
- Curve C must be traversed **counterclockwise** (positive orientation)
- P and Q must have **continuous partial derivatives** in D
- D must be a **simply connected** region (no holes)

7.4 Special Case: Area Using Green's Theorem

$$\text{Area of } D = (1/2) \oint_C (x \, dy - y \, dx)$$

This is derived by choosing $P = -y/2$, $Q = x/2$, giving $\partial Q / \partial x - \partial P / \partial y = 1/2 + 1/2 = 1$.

7.5 Green's Theorem in Vector Form

Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$. Then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA = \iint_D (\partial Q / \partial x - \partial P / \partial y) \, dA$$

Applications of Green's Theorem

8.1 Computing Line Integrals via Double Integrals

When direct line integral computation is complex (curved path, complicated parametrization), Green's theorem allows us to convert it to a double integral over the region. This is often simpler.

8.2 Computing Area of a Region

Green's theorem gives us three equivalent area formulas:

$$\text{Area} = \oint_C x \, dy = -\oint_C y \, dx = (1/2) \oint_C (x \, dy - y \, dx)$$

These are derived by substituting specific values of P and Q into Green's theorem. This is especially useful for regions bounded by curves where direct area integration would be complicated.

8.3 Verification Problems

Green's theorem is often tested by asking you to verify it — compute both sides (line integral and double integral) and show they are equal.

8.4 Converting Hard Line Integrals to Easy Double Integrals

Example situation: If you're asked to compute $\oint_C [(x^2 - y^2) \, dx + 2xy \, dy]$ around a complicated closed curve, Green's theorem converts this to $\iint_D 4y \, dA$, which might be much easier to compute directly.

8.5 Limitations

Common Mistake: Green's Theorem applies ONLY in 2D (the xy-plane). For 3D surfaces, we must use Stokes' Theorem instead.

Common Mistake: If the curve is traversed clockwise, a negative sign must be introduced: $\oint_{-C} (P \, dx + Q \, dy) = -\iint_D (\partial Q/\partial x - \partial P/\partial y) \, dA$

Stokes' Theorem (Without Proof)

9.1 Statement of Stokes' Theorem

Stokes' Theorem

Let S be an oriented, piecewise-smooth surface bounded by a simple, closed, piecewise-smooth curve C . Let $\mathbf{F}(x,y,z)$ be a vector field with continuous first partial derivatives on an open region containing S . Then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$$

The orientation of C and S must be compatible (right-hand rule: if thumb points in direction of $\mathbf{n}$, fingers curl in direction of positive traversal of C).

9.2 Intuitive Meaning

Stokes' theorem is the 3D generalization of Green's theorem. It says:

- The **circulation** of a vector field around a closed loop C equals the **flux of the curl** through any surface S bounded by C
- This means we can choose ANY surface bounded by C — a flat surface or a curved one — the answer will be the same!

9.3 Computing curl $\mathbf{F}$

$$\begin{aligned} \text{curl } \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix} \\ &= (\partial R/\partial y - \partial Q/\partial z) \mathbf{i} - (\partial R/\partial x - \partial P/\partial z) \mathbf{j} + (\partial Q/\partial x - \partial P/\partial y) \mathbf{k} \end{aligned}$$

9.4 Conditions for Stokes' Theorem

- S must be an **oriented, smooth (or piecewise-smooth)** surface
- C must be the **positively oriented boundary** of S (by right-hand rule)
- $\mathbf{F}$ must have **continuous first-order partial derivatives** on an open set containing S

9.5 Relationship to Green's Theorem

If S is a flat surface in the xy -plane (so $\mathbf{n} = \mathbf{k}$), then Stokes' theorem reduces exactly to Green's theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k} \, dA = \iint_D (\partial Q / \partial x - \partial P / \partial y) \, dA$$

So Green's Theorem is a special case of Stokes' Theorem in 2D.

Applications of Stokes' Theorem

10.1 Computing Line Integrals via Surface Integrals

When the line integral over a closed curve in 3D is hard to compute directly, Stokes' theorem converts it to a surface integral of $\text{curl } \mathbf{F}$, which may be simpler — especially if $\text{curl } \mathbf{F}$ is simple or if we can choose a flat surface.

10.2 Proving a Field is Conservative

If $\text{curl } \mathbf{F} = \mathbf{0}$ everywhere, then by Stokes' theorem, $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for all closed curves C . This means $\mathbf{F}$ is conservative.

10.3 Applications in Electromagnetism

- **Faraday's Law:** EMF around a loop $= -d\Phi_B/dt$ relates the circulation of $\mathbf{E}$ to the flux of $\mathbf{B}$ (via Stokes' theorem)
- **Ampere's Law:** Magnetic field circulation around a loop relates to current through the surface

10.4 Choosing the Best Surface

A key advantage of Stokes' theorem: we can choose any surface bounded by C . We typically choose:

- A **flat (plane) surface** — simplest; normal is constant
- A surface where **$\text{curl } \mathbf{F}$ is zero** on most of it — simplifies computation
- A surface that makes the **integral limits simple**

Note: Always check: if $\text{curl } \mathbf{F} = 0$ throughout, the line integral is zero without any computation!

Volume Integrals

11.1 What is a Volume Integral?

A **volume integral** integrates a scalar or vector function over a three-dimensional region V in space. It generalizes the concept of double integrals to 3D.

Volume Integral of a Scalar $f(x,y,z)$ over region V :

$$\iiint_V f(x,y,z) \, dV = \iiint_V f(x,y,z) \, dx \, dy \, dz$$

Volume Integral of a Vector $\mathbf{F}(x,y,z)$ over region V :

$$\iiint_V \mathbf{F} \, dV = \iiint_V P \, dV \mathbf{i} + \iiint_V Q \, dV \mathbf{j} + \iiint_V R \, dV \mathbf{k}$$

11.2 Physical Applications of Volume Integrals

Quantity	Formula	$f(x,y,z)$
Volume of region V	$\iiint_V dV$	$f = 1$
Total mass (density ρ)	$\iiint_V \rho \, dV$	$f = \text{density } \rho$
Total charge (charge density ρ)	$\iiint_V \rho \, dV$	$f = \text{charge density}$
Moment of inertia (about z-axis)	$\iiint_V (x^2 + y^2) \rho \, dV$	$f = (x^2 + y^2)\rho$
Average value of f	$(1/V) \iiint_V f \, dV$	Any f

11.3 Evaluating Volume Integrals

Evaluate by converting to iterated integrals (Cartesian, cylindrical, or spherical). The choice of coordinate system depends on the shape of V :

- **Box/rectangular region:** Use Cartesian (x,y,z)
- **Cylinder:** Use cylindrical (r, θ, z) where $x = r \cos\theta$, $y = r \sin\theta$, $z = z$

- **Sphere:** Use spherical (ρ, ϕ, θ) where $x = \rho \sin\phi \cos\theta$, $y = \rho \sin\phi \sin\theta$, $z = \rho \cos\phi$

$$\text{Cylindrical: } dV = r \, dr \, d\theta \, dz$$

$$\text{Spherical: } dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Divergence Theorem (Without Proof)

12.1 Statement of the Divergence Theorem

Gauss's Divergence Theorem

Let V be a closed bounded region in 3D space with boundary surface S (a closed surface). Let $\mathbf{F}(x,y,z)$ be a vector field with continuous first partial derivatives on an open region containing V . Then:

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

where $\mathbf{n}$ is the outward unit normal to S , and $\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z$.

12.2 Physical Interpretation

The Divergence Theorem says: *The total outward flux of a vector field through a closed surface S equals the integral of the divergence of the field throughout the enclosed volume V .*

- If $\text{div } \mathbf{F} > 0$ at a point: that point is a **source** (field lines emanate outward)
- If $\text{div } \mathbf{F} < 0$ at a point: that point is a **sink** (field lines converge inward)
- If $\text{div } \mathbf{F} = 0$ everywhere: no sources or sinks (incompressible flow)

12.3 Computing Divergence

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z$$

Example: If $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$, then $\text{div } \mathbf{F} = 2x + 2y + 2z$.

12.4 Summary of All Three Theorems

Theorem	Dimension	Converts	Key Quantity
Green's Theorem	2D	Line integral $\rightarrow$ Double integral	$\partial Q/\partial x - \partial P/\partial y$
Stokes' Theorem	3D	Line integral $\rightarrow$ Surface integral	$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$

Divergence Theorem	3D	Surface integral $\rightarrow$ Volume integral	$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$
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Applications of Divergence Theorem

13.1 Computing Surface Integrals via Volume Integrals

When computing the flux through a closed surface directly is complex (many faces, curved surfaces), the Divergence Theorem converts it to a volume integral, which may be much easier — especially if $\text{div } \mathbf{F}$ is simple.

13.2 Gauss's Law in Electrostatics

The most famous application: For electric field $\mathbf{E}$ due to a charge distribution,

$$\iint_S \mathbf{E} \cdot \mathbf{n} \, dS = Q_{\text{enc}} / \epsilon_0$$

This follows from $\text{div } \mathbf{E} = \rho / \epsilon_0$ (a Maxwell equation) and the Divergence Theorem.

13.3 Continuity Equation in Fluid Mechanics

For an incompressible fluid with velocity field $\mathbf{v}$:

$$\text{div } \mathbf{v} = 0 \quad \Rightarrow \quad \iint_S \mathbf{v} \cdot \mathbf{n} \, dS = 0$$

This means the net flow out of any closed volume is zero — no fluid is created or destroyed.

13.4 Verifying the Divergence Theorem

A classic exam problem: "Verify the Divergence Theorem for $\mathbf{F} = \dots$ over the cube/sphere/cylinder ..."

Steps: (1) Compute the surface integral (left side) face-by-face, (2) Compute the volume integral (right side), (3) Show both sides are equal.

Solved Examples (40+ Ultra-Detailed)

Example 1 – Basic Line Integral Along a Straight Line

Problem: Evaluate $\int_C (x^2 + y) dx + (y - x) dy$ where C is the straight line segment from $(0,0)$ to $(1,1)$.

Step 1: Identify the Problem

We have a line integral $\int_C P dx + Q dy$ where $P = x^2 + y$ and $Q = y - x$. The curve C is a straight line from $(0,0)$ to $(1,1)$.

Step 2: Parametrize the Curve

For a straight line from $(0,0)$ to $(1,1)$, use parameter t from 0 to 1:

$$x = 0 + t(1-0) = t$$

$$y = 0 + t(1-0) = t$$

Note: along this line, $y = x$ (both equal t).

Step 3: Compute Differentials

$$dx/dt = 1, \text{ so } dx = dt$$

$$dy/dt = 1, \text{ so } dy = dt$$

Step 4: Set Limits

When $t = 0$: $(x,y) = (0,0)$ = starting point ✓

When $t = 1$: $(x,y) = (1,1)$ = ending point ✓

So t ranges from 0 to 1.

Step 5: Substitute into Integral

Replace $x = t$, $y = t$, $dx = dt$, $dy = dt$:

$$P = x^2 + y = t^2 + t$$

$$Q = y - x = t - t = 0$$

$$\text{Integral becomes: } \int_0^1 [(t^2+t) \cdot dt + 0 \cdot dt] = \int_0^1 (t^2+t) dt$$

Step 6: Perform Integration

$$\begin{aligned}\int_0^1 (t^2+t) dt &= [t^3/3 + t^2/2]_0^1 \\ &= (1/3 + 1/2) - (0 + 0) \\ &= 2/6 + 3/6 = 5/6\end{aligned}$$

Answer: 5/6

Physical Interpretation: This represents the "work" done by the force field $F = (x^2+y)i + (y-x)j$ as a particle moves from $(0,0)$ to $(1,1)$ along the given path.

Example 2 – Line Integral Along a Circle

Problem: Evaluate $\int_C (x dx + y dy)$ where C is the circle $x^2 + y^2 = 4$ traversed counterclockwise.

Step 1: Identify P and Q

Here $P = x$ and $Q = y$. The curve is a circle of radius 2 centered at origin.

Step 2: Check if Conservative (Optional Check)

$\partial Q/\partial x = 0$ and $\partial P/\partial y = 0$. Since they are equal, $F = x i + y j$ is a conservative field. For a closed path, the integral of a conservative field is ZERO!

Step 3: Parametrize for Verification

$$\begin{aligned}x &= 2 \cos t, y = 2 \sin t, t: 0 \text{ to } 2\pi \\ dx &= -2 \sin t dt, dy = 2 \cos t dt\end{aligned}$$

Step 4: Substitute and Integrate

$$\begin{aligned}\int_0^{2\pi} [(2 \cos t)(-2 \sin t) + (2 \sin t)(2 \cos t)] dt \\ &= \int_0^{2\pi} [-4 \sin t \cos t + 4 \sin t \cos t] dt \\ &= \int_0^{2\pi} 0 dt = 0\end{aligned}$$

Answer: 0

Physical Interpretation: Since F is conservative (gradient of $x^2/2 + y^2/2$), no net work is done around any closed path.

Example 3 – Work Done Along a Parabolic Path

Problem: Find the work done by $F = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ along the parabola $y = x^2$ from $(0,0)$ to $(1,1)$.

Step 1: State the Problem

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (x^2 - y^2)dx + 2xy dy, \text{ along } y = x^2.$$

Step 2: Parametrize the Parabola

On the parabola $y = x^2$, use $x = t$ as parameter:

$$x = t, y = t^2$$

$$dx = dt, dy = 2t dt$$

t goes from 0 (at origin) to 1 (at $(1,1)$)

Step 3: Substitute

$$x^2 - y^2 = t^2 - t^4$$

$$2xy = 2t \cdot t^2 = 2t^3$$

$$W = \int_0^1 [(t^2 - t^4)(1) + (2t^3)(2t)] dt$$

$$= \int_0^1 (t^2 - t^4 + 4t^4) dt$$

$$= \int_0^1 (t^2 + 3t^4) dt$$

Step 4: Integrate Term by Term

$$= [t^3/3 + 3t^5/5]_0^1$$

$$= (1/3 + 3/5) - 0$$

$$= 5/15 + 9/15 = 14/15$$

Answer: $W = 14/15$ units of work

Physical Interpretation: The force field does $14/15$ units of work in moving the particle from $(0,0)$ to $(1,1)$ along the parabolic path $y = x^2$.

Example 4 – Line Integral Along a Broken Path

Problem: Evaluate $\int_C xy \, dx + x^2 \, dy$ where C consists of (i) line from $(0,0)$ to $(1,0)$, then (ii) line from $(1,0)$ to $(1,2)$.

Step 1: Identify the Segments

$$C = C_1 + C_2. \text{ Total integral} = \int_{C_1} + \int_{C_2}.$$

Step 2: Segment C_1 — from $(0,0)$ to $(1,0)$

This is along the x-axis, so $y = 0$, $dy = 0$.

x goes from 0 to 1: parametrize as $x = t$, $y = 0$, $t: 0$ to 1.

$$\int_{C_1} = \int_0^1 [(t)(0)(1) + t^2(0)] \, dt = \int_0^1 0 \, dt = 0$$

Step 3: Segment C_2 — from $(1,0)$ to $(1,2)$

This is along $x = 1$ (vertical line), so $x = 1$, $dx = 0$.

y goes from 0 to 2: parametrize as $x = 1$, $y = t$, $t: 0$ to 2.

$$\int_{C_2} = \int_0^2 [(1)(t)(0) + 1^2(1)] \, dt = \int_0^2 1 \, dt = [t]_0^2 = 2$$

Step 4: Total Integral

$$\int_C = 0 + 2 = 2$$

Answer: 2

Example 5 – Green's Theorem Verification

Problem: Verify Green's Theorem for $F = (x^2 - y^2)\mathbf{i} + 2xy \, \mathbf{j}$ over the square with vertices $(0,0)$, $(2,0)$, $(2,2)$, $(0,2)$ traversed counterclockwise.

Step 1: Identify P, Q and the Region D

$P = x^2 - y^2$, $Q = 2xy$. D is the square $0 \leq x \leq 2$, $0 \leq y \leq 2$.

Step 2: Compute RIGHT SIDE (Double Integral)

$$\partial Q / \partial x = 2y, \partial P / \partial y = -2y$$

$$\partial Q / \partial x - \partial P / \partial y = 2y - (-2y) = 4y$$

$$\begin{aligned} \iint_D 4y \, dA &= \int_0^2 \int_0^2 4y \, dy \, dx \\ &= \int_0^2 [2y^2]_0^2 \, dx = \int_0^2 8 \, dx = 8[x]_0^2 = 16 \end{aligned}$$

Step 3: Compute LEFT SIDE (Line Integral) — 4 sides

Side 1: Bottom: $y=0$, $dy=0$, x : 0 to 2. $\int = \int_0^2 (x^2-0)dx + 0 = [x^3/3]_0^2 = 8/3$

Side 2: Right: $x=2$, $dx=0$, y : 0 to 2. $\int = 0 + \int_0^2 2(2)y \, dy = 4[y^2/2]_0^2 = 8$

Side 3: Top: $y=2$, $dy=0$, x : 2 to 0. $\int = \int_2^0 (x^2-4)dx = [x^3/3 - 4x]_2^0 = 0 - (8/3 - 8) = -8/3 + 8 = 16/3$

Side 4: Left: $x=0$, $dx=0$, y : 2 to 0. $\int = 0 + \int_2^0 0 \, dy = 0$

Total = $8/3 + 8 + 16/3 + 0 = 24/3 + 8 = 8 + 8 = 16$

Step 4: Verify Both Sides Equal

Left side = 16 = Right side. Green's Theorem verified! ✓

Both sides = 16. Green's Theorem Verified.

Example 6 – Area Using Green's Theorem

Problem: Find the area enclosed by the ellipse $x^2/a^2 + y^2/b^2 = 1$ using Green's theorem.

Step 1: Apply the Area Formula

Area = $(1/2) \oint_C (x \, dy - y \, dx)$, where C is the ellipse traversed counterclockwise.

Step 2: Parametrize the Ellipse

$$x = a \cos t, y = b \sin t, t: 0 \text{ to } 2\pi$$

$$dx = -a \sin t \, dt$$

$$dy = b \cos t \, dt$$

Step 3: Compute $x \, dy - y \, dx$

$$x \, dy = (a \cos t)(b \cos t \, dt) = ab \cos^2 t \, dt$$

$$y \, dx = (b \sin t)(-a \sin t \, dt) = -ab \sin^2 t \, dt$$

$$x \, dy - y \, dx = ab \cos^2 t \, dt + ab \sin^2 t \, dt = ab(\cos^2 t + \sin^2 t) \, dt = ab \, dt$$

Step 4: Compute the Area

$$\text{Area} = (1/2) \int_0^{2\pi} ab \, dt = (1/2)(ab)(2\pi) = \pi ab$$

$$\text{Area of Ellipse} = \pi ab$$

Physical Interpretation: This is the well-known formula for the area of an ellipse with semi-axes a and b . Green's theorem derives it elegantly from the boundary curve alone.

Example 7 – Using Green's Theorem to Evaluate a Line Integral

Problem: Use Green's Theorem to evaluate $\oint_C (3y - e^{\sin x}) \, dx + (7x + \sqrt{y^4+1}) \, dy$ where C is the circle $x^2+y^2=9$, counterclockwise.

Step 1: Identify P and Q

$$P = 3y - e^{\sin x}, Q = 7x + \sqrt{y^4+1}$$

Step 2: Compute Partial Derivatives

$$\partial Q / \partial x = 7 \text{ (derivative of } 7x \text{ with respect to } x; \sqrt{y^4+1} \text{ has no } x)$$

$$\partial P / \partial y = 3 \text{ (derivative of } 3y \text{ with respect to } y; e^{\sin x} \text{ has no } y)$$

$$\partial Q / \partial x - \partial P / \partial y = 7 - 3 = 4$$

Step 3: Apply Green's Theorem

$$\oint_C (...) = \iint_D 4 \, dA, \text{ where } D \text{ is the disk } x^2+y^2 \leq 9 \text{ (radius 3)}$$

Step 4: Compute the Double Integral

$$\iint_D 4 \, dA = 4 \times \text{Area of disk} = 4 \times \pi(3)^2 = 4 \times 9\pi = 36\pi$$

Answer: 36π

Physical Interpretation: Green's Theorem simplified a near-impossible line integral (involving $e^{\sin x}$ and $\sqrt{(y^4+1)}$) into a trivial area calculation!

Example 8 – Line Integral Along a Helix (3D)

Problem: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = y \mathbf{i} - x \mathbf{j} + z \mathbf{k}$ and C is the helix $x = \cos t$, $y = \sin t$, $z = t$, $t: 0$ to 2π .

Step 1: Identify the Setup

We need $\int_C (y \, dx - x \, dy + z \, dz)$. The curve is a helix on a cylinder of radius 1, rising by 2π in one full revolution.

Step 2: Compute Differentials

$$x = \cos t \Rightarrow dx = -\sin t \, dt$$

$$y = \sin t \Rightarrow dy = \cos t \, dt$$

$$z = t \Rightarrow dz = dt$$

Step 3: Substitute into $\mathbf{F} \cdot d\mathbf{r}$

$$y \, dx = (\sin t)(-\sin t \, dt) = -\sin^2 t \, dt$$

$$-x \, dy = -(\cos t)(\cos t \, dt) = -\cos^2 t \, dt$$

$$z \, dz = t \, dt$$

$$\mathbf{F} \cdot d\mathbf{r} = (-\sin^2 t - \cos^2 t + t) \, dt = (-1 + t) \, dt$$

Step 4: Integrate

$$\int_0^{2\pi} (-1 + t) \, dt = [-t + t^2/2]_0^{2\pi}$$

$$= -2\pi + (2\pi)^2/2 = -2\pi + 2\pi^2 = 2\pi(\pi - 1)$$

Answer: $2\pi(\pi - 1)$

Example 9 – Stokes' Theorem Application

Problem: Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = z^2\mathbf{i} + x^2\mathbf{j} + y^2\mathbf{k}$ and C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ traversed counterclockwise when viewed from above.

Step 1: Find curl $\mathbf{F}$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z^2 & x^2 & y^2 \end{vmatrix}$$

$$\text{i-component: } \partial(y^2)/\partial y - \partial(x^2)/\partial z = 2y - 0 = 2y$$

$$\text{j-component: } -[\partial(y^2)/\partial x - \partial(z^2)/\partial z] = -[0 - 2z] = 2z$$

$$\text{k-component: } \partial(x^2)/\partial x - \partial(z^2)/\partial y = 2x - 0 = 2x$$

$$\text{curl } \mathbf{F} = 2y \mathbf{i} + 2z \mathbf{j} + 2x \mathbf{k}$$

Step 2: Identify the Surface S

The boundary C is the triangle. Use the triangular plane surface S in the plane $x + y + z = 1$ (verify: all three vertices satisfy this).

Step 3: Find Normal to Surface $x+y+z=1$

The plane $x + y + z = 1$ has equation $F(x,y,z) = x+y+z-1 = 0$.

$$\nabla F = \mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$|\nabla F| = \sqrt{3}$$

$$\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$$

Step 4: Compute $(\text{curl } \mathbf{F}) \cdot \mathbf{n}$

$$(\text{curl } \mathbf{F}) \cdot \mathbf{n} = (2y \mathbf{i} + 2z \mathbf{j} + 2x \mathbf{k}) \cdot (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$$

$$= (2y + 2z + 2x)/\sqrt{3} = 2(x+y+z)/\sqrt{3}$$

$$\text{On the surface } x+y+z = 1: = 2/\sqrt{3}$$

Step 5: Compute dS and the Integral

dS (area element) for projection onto xy -plane: $dS = \sqrt{3} \, dA$ (since $|\nabla F|/|\partial F/\partial z| = \sqrt{3}/1 = \sqrt{3}$)

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_D (2/\sqrt{3})(\sqrt{3}) \, dA = \iint_D 2 \, dA$$

D is the projection of triangle onto xy -plane: triangle with vertices $(1,0)$, $(0,1)$, $(0,0)$. Area = $1/2$.

$$= 2 \times (1/2) = 1$$

Answer: 1

Physical Interpretation: The circulation of the swirling force field F around the triangular loop equals 1 unit.

Example 10 – Divergence Theorem: Flux Through a Cube

Problem: Use the Divergence Theorem to find the flux of $F = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$ outward through the surface of the cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.

Step 1: Why Use Divergence Theorem?

The cube has 6 faces. Computing flux through each face separately would require 6 integrals. Divergence Theorem converts this to ONE triple integral. Much easier!

Step 2: Compute $\text{div } F$

$$\text{div } F = \partial(x^3)/\partial x + \partial(y^3)/\partial y + \partial(z^3)/\partial z = 3x^2 + 3y^2 + 3z^2$$

Step 3: Set Up Volume Integral

$$\begin{aligned}\text{Flux} &= \iiint_V (3x^2 + 3y^2 + 3z^2) \, dV \\ &= \int_0^1 \int_0^1 \int_0^1 (3x^2 + 3y^2 + 3z^2) \, dz \, dy \, dx\end{aligned}$$

Step 4: Integrate Step by Step

$$\text{Inner (z): } \int_0^1 (3x^2 + 3y^2 + 3z^2) \, dz = [3x^2z + 3y^2z + z^3]_0^1 = 3x^2 + 3y^2 + 1$$

$$\text{Middle (y): } \int_0^1 (3x^2 + 3y^2 + 1) \, dy = [3x^2y + y^3 + y]_0^1 = 3x^2 + 1 + 1 = 3x^2 + 2$$

$$\text{Outer (x): } \int_0^1 (3x^2 + 2) \, dx = [x^3 + 2x]_0^1 = 1 + 2 = 3$$

Total Flux = 3

Physical Interpretation: 3 units of the vector field "flow out" of the unit cube per unit time — representing, for example, 3 units of fluid outflow.

Example 11 – Flux Through a Sphere Using Divergence Theorem

Problem: Find the flux of $F = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ outward through the sphere $x^2 + y^2 + z^2 = a^2$.

Step 1: Compute $\text{div } F$

$$\text{div } F = \partial x / \partial x + \partial y / \partial y + \partial z / \partial z = 1 + 1 + 1 = 3$$

Step 2: Apply Divergence Theorem

$$\text{Flux} = \iiint_V 3 \, dV = 3 \times \text{Volume of sphere}$$

Step 3: Volume of Sphere

$$V = (4/3)\pi a^3$$

$$\text{Flux} = 3 \times (4/3)\pi a^3 = 4\pi a^3$$

$$\text{Flux} = 4\pi a^3$$

Physical Interpretation: The position field $F = r$ has constant divergence 3. The total outward flux through a sphere of radius a equals $4\pi a^3 = 3 \times \text{volume of sphere}$.

Example 12 – Surface Integral Over a Plane

Problem: Evaluate $\iint_S F \cdot dS$ where $F = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ and S is the portion of the plane $z = 1$ inside the cylinder $x^2 + y^2 = 4$, with normal pointing upward.

Step 1: Identify the Surface

$$S: z = 1 \text{ (constant), so } \partial z / \partial x = 0, \partial z / \partial y = 0. \text{ Normal pointing upward: } dS = \mathbf{k} \, dx \, dy.$$

Step 2: Compute $F \cdot dS$

$F \cdot k = z = 1$ (on the surface $z=1$)

So $F \cdot dS = 1 \, dx \, dy$

Step 3: Set Up Integral

$\iint_S F \cdot dS = \iint_D 1 \, dA$ where D is the disk $x^2 + y^2 \leq 4$

Step 4: Compute Area of Disk

Area $= \pi r^2 = \pi(2)^2 = 4\pi$

$\iint_S F \cdot dS = 4\pi$

$$\text{Flux} = 4\pi$$

Example 13 – Stokes' Theorem: Curl Integral = 0

Problem: Evaluate $\oint_C F \cdot dr$ where $F = \nabla(x^2y + e^z)$ and C is any closed curve.

Step 1: Recognize F is a Gradient Field

$F = \nabla\phi$ where $\phi = x^2y + e^z$. A gradient field is always conservative.

Step 2: Compute curl F

For any gradient field $\nabla\phi$, $\text{curl}(\nabla\phi) = 0$ (this is a standard identity).

Step 3: Apply Stokes' Theorem

$\oint_C F \cdot dr = \iint_S (\text{curl } F) \cdot dS = \iint_S 0 \, dS = 0$

Answer: 0 (for any closed curve C)

Physical Interpretation: A conservative field does zero net work around any closed path — energy is conserved.

Example 14 – Volume Integral (Total Mass)

Problem: Find the total mass of the cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$ if density $\rho(x,y,z) = xyz$.

Step 1: Set Up the Volume Integral

$$\text{Mass} = \iiint_V \rho \, dV = \int_0^1 \int_0^1 \int_0^1 xyz \, dz \, dy \, dx$$

Step 2: Inner Integral (z)

$$\int_0^1 xyz \, dz = xy \left[\frac{z^2}{2} \right]_0^1 = xy/2$$

Step 3: Middle Integral (y)

$$\int_0^1 (xy/2) \, dy = (x/2) \left[\frac{y^2}{2} \right]_0^1 = x/4$$

Step 4: Outer Integral (x)

$$\int_0^1 (x/4) \, dx = (1/4) \left[\frac{x^2}{2} \right]_0^1 = 1/8$$

Total Mass = 1/8 units

Example 15 – Divergence Theorem Verification

Problem: Verify the Divergence Theorem for $F = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$ over the cylinder $x^2 + y^2 \leq 1$, $0 \leq z \leq 1$.

Step 1: Compute div F (for Volume Integral)

$$\text{div } F = 2x + 2y + 2z$$

Step 2: Compute Volume Integral (Right Side)

Use cylindrical coordinates: $x = r \cos\theta$, $y = r \sin\theta$, $z = z$, $dV = r \, dr \, d\theta \, dz$

$$\begin{aligned} \iiint_V 2(x+y+z) \, dV &= \int_0^{2\pi} \int_0^1 \int_0^1 2(r \cos\theta + r \sin\theta + z) r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 \int_0^1 [2r^2(\cos\theta + \sin\theta) + 2rz] \, dz \, dr \, d\theta \end{aligned}$$

Note: $\int_0^{2\pi} (\cos\theta + \sin\theta) \, d\theta = 0$, so first term vanishes.

$$\text{Remaining: } \int_0^{2\pi} \int_0^1 \int_0^1 2rz \, dz \, dr \, d\theta$$

Inner: $\int_0^1 2rz \, dz = r[z^2]_0^1 = r$

Middle: $\int_0^1 r \, dr = 1/2$

Outer: $\int_0^{2\pi} (1/2) \, d\theta = \pi$

Step 3: Surface Integral (Left Side — 3 parts)

Top (z=1, n=k): $F \cdot k = z^2 = 1$; $\iint = \iint_D 1 \, dA = \pi(1)^2 = \pi$

Bottom (z=0, n=-k): $F \cdot (-k) = -z^2 = 0$; $\iint = 0$

Lateral (x²+y²=1, n=(x i+y j)): $F \cdot n = x^2 \cdot x + y^2 \cdot y = x^3 + y^3$
 $= \int_0^{2\pi} \int_0^1 (\cos^3 \theta + \sin^3 \theta) \, dz \, d\theta = 0$ (by symmetry over full period)

Total Surface Integral = $\pi + 0 + 0 = \pi$

Step 4: Verify

Volume integral = π = Surface integral. ✓

Both sides = π . Divergence Theorem Verified.

Example 16 – Line Integral Along $y = x^2$ (another path)

Problem: Evaluate $\int_C y^2 \, dx + xy \, dy$ from (0,0) to (2,4) along $y = x^2$.

Step 1: Parametrize

$x = t, y = t^2, dx = dt, dy = 2t \, dt$; t : 0 to 2.

Step 2: Substitute

$y^2 \, dx + xy \, dy = (t^4)(1) + (t)(t^2)(2t) \, dt = (t^4 + 2t^4) \, dt = 3t^4 \, dt$

Step 3: Integrate

$\int_0^2 3t^4 \, dt = [3t^5/5]_0^2 = 3(32)/5 = 96/5$

Answer: 96/5

Example 17 – Circulation Around a Triangle

Problem: Find $\oint_C (x+y) dx + (x-y) dy$ around the triangle $(0,0)$, $(1,0)$, $(0,1)$ counterclockwise.

Step 1: Use Green's Theorem

$$P = x+y, Q = x-y. \partial Q/\partial x - \partial P/\partial y = 1 - 1 = 0.$$

Step 2: Apply

$$\oint_C = \iint_D 0 \, dA = 0.$$

Answer: 0 (No net circulation)

Example 18 – Work Done by $F = y\mathbf{i} + x\mathbf{j}$

Problem: Find work done by $F = y\mathbf{i} + x\mathbf{j}$ from $(0,0)$ to $(1,1)$ along (a) straight line, (b) $y=x^2$.

Step 1: Check Conservatism

$$\partial Q/\partial x = 1 = \partial P/\partial y = 1. \text{ So } F \text{ is conservative! Work} = \varphi(1,1) - \varphi(0,0).$$

Step 2: Find Potential

$$F = \nabla\varphi \text{ means } \partial\varphi/\partial x = y \text{ and } \partial\varphi/\partial y = x. \text{ So } \varphi = xy.$$

Step 3: Compute

$$W = \varphi(1,1) - \varphi(0,0) = 1 - 0 = 1 \text{ (same for BOTH paths!)}$$

$W = 1$ (path-independent)

Physical Interpretation: Since F is conservative, work done is independent of the path taken — only the endpoints matter.

Example 19 – Surface Integral Over a Tilted Plane

Problem: Evaluate $\iint_S (x+y+z) \, dS$ where S is the triangle in the plane $x+y+z=1$ in the first octant.

Step 1: Parametrize

On S : $z = 1-x-y$. Region D : $x \geq 0, y \geq 0, x+y \leq 1$.

Step 2: Compute dS

$g_x = -1, g_y = -1$. $dS = \sqrt{(1+1+1)} \, dA = \sqrt{3} \, dA$

Step 3: On the surface

$x+y+z = 1$ on S . So integrand $= 1$.

Step 4: Compute

$\iint_S 1 \cdot \sqrt{3} \, dA = \sqrt{3} \times \text{Area}(D) = \sqrt{3} \times (1/2) = \sqrt{3}/2$

Answer: $\sqrt{3}/2$

Example 20 – Divergence Theorem: Flux Through a Cone + Disk

Problem: Find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = x \mathbf{i} + y \mathbf{j} + z^2 \mathbf{k}$ and S is the closed surface formed by $z = \sqrt{x^2+y^2}$ and $z=1$ (the cap).

Step 1: Compute $\text{div } \mathbf{F}$

$\text{div } \mathbf{F} = 1 + 1 + 2z = 2 + 2z$

Step 2: Set up in Cylindrical Coords

The region V : $0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, r \leq z \leq 1$ (cone up to cap).

Step 3: Volume Integral

$$\iiint_V (2+2z) r \, dz \, dr \, d\theta$$

$$\text{Inner: } \int_r^1 (2+2z)r \, dz = r[2z+z^2]_r^1 = r(3 - 2r - r^2)$$

$$\text{Middle: } \int_0^1 r(3-2r-r^2) \, dr = \int_0^1 (3r-2r^2-r^3) \, dr = [3r^2/2 - 2r^3/3 - r^4/4]_0^1 = 3/2 - 2/3 - 1/4 = 18/12 - 8/12 - 3/12 = 7/12$$

$$\text{Outer: } \int_0^{2\pi} (7/12) \, d\theta = 7\pi/6$$

$$\text{Flux} = 7\pi/6$$

Example 21 – Line Integral of a Scalar Function

Problem: Evaluate $\int_C (x^2+y^2) \, ds$ along the semicircle $x=\cos t$, $y=\sin t$, $t: 0$ to π .

Step 1: Compute ds

$$ds = \sqrt{((dx/dt)^2 + (dy/dt)^2)} \, dt = \sqrt{(\sin^2 t + \cos^2 t)} \, dt = dt$$

Step 2: On the circle

$$x^2+y^2 = \cos^2 t + \sin^2 t = 1$$

Step 3: Integrate

$$\int_0^\pi 1 \, dt = \pi$$

$$\text{Answer: } \pi$$

Example 22 – Stokes' Theorem on a Hemisphere

Problem: Verify Stokes' theorem for $F = -y \, \mathbf{i} + x \, \mathbf{j} + z \, \mathbf{k}$ on the upper hemisphere $z = \sqrt{4-x^2-y^2}$, bounded by circle $x^2+y^2=4$, $z=0$.

Step 1: Line Integral (Left Side)

C: circle $x=2\cos t$, $y=2\sin t$, $z=0$, $t: 0$ to 2π

$$dx = -2\sin t \, dt, \, dy = 2\cos t \, dt$$

$$F = (-2\sin t, 2\cos t, 0)$$

$$F \cdot dr = (-2\sin t)(-2\sin t \, dt) + (2\cos t)(2\cos t \, dt) = 4\sin^2 t \, dt + 4\cos^2 t \, dt = 4 \, dt$$

$$\oint_C = \int_0^{2\pi} 4 \, dt = 8\pi$$

Step 2: Curl F

$$\text{curl } F = (0-0)\mathbf{i} - (0-0)\mathbf{j} + (1-(-1))\mathbf{k} = 2\mathbf{k}$$

Step 3: Surface Integral (Right Side)

Choose flat disk $S: z=0, x^2+y^2 \leq 4$; $\mathbf{n}=\mathbf{k}$ (upward)

$$\iint_S \text{curl } F \cdot d\mathbf{S} = \iint_D 2\mathbf{k} \cdot \mathbf{k} \, dA = \iint_D 2 \, dA = 2 \times \pi(4) = 8\pi$$

Step 4: Verify

Both sides = 8π ✓

Stokes' Theorem Verified: Both sides = 8π

Example 23 – Green's Theorem: Converting Complex Line Integral

Problem: Evaluate $\oint_C (x^3 - y^3) \, dx + (x^3 + y^3) \, dy$ around the circle $x^2+y^2=4$.

Step 1

$$P = x^3 - y^3, \, Q = x^3 + y^3$$

$$\partial Q / \partial x - \partial P / \partial y = 3x^2 - (-3y^2) = 3x^2 + 3y^2 = 3(x^2 + y^2)$$

Step 2: In polar

$$\iint_D 3(x^2 + y^2) \, dA = \int_0^{2\pi} \int_0^2 3r^2 \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^2 3r^3 \, dr \, d\theta$$

$$\text{Inner: } [3r^4/4]_0^2 = 12$$

$$\text{Outer: } \int_0^{2\pi} 12 \, d\theta = 24\pi$$

Answer: 24π

Example 24 – Volume Integral in Spherical Coordinates

Problem: Evaluate $\iiint_V (x^2+y^2+z^2) \, dV$ over the sphere of radius a .

Step 1: Use Spherical Coordinates

$$x^2+y^2+z^2 = \rho^2, \, dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Step 2: Set Up

$$\iiint_V \rho^2 \cdot \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^\pi \int_0^a \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$$

Step 3: Integrate

$$\int_0^a \rho^4 \, d\rho = a^5/5$$

$$\int_0^\pi \sin\phi \, d\phi = 2$$

$$\int_0^{2\pi} d\theta = 2\pi$$

$$\text{Total} = (a^5/5)(2)(2\pi) = 4\pi a^5/5$$

Answer: $4\pi a^5/5$

Example 25 – Work Done Around a Closed Path: Non-Conservative

Problem: Find $\oint_C (-y \, dx + x \, dy)$ around the circle $x^2+y^2=r^2$, counterclockwise.

Step 1: Use Green's Theorem

$$P=-y, \, Q=x. \, \partial Q/\partial x - \partial P/\partial y = 1-(-1) = 2.$$

$$\iint_D 2 \, dA = 2 \times \pi r^2 = 2\pi r^2$$

$$\text{Circulation} = 2\pi r^2$$

Physical Interpretation: This is the well-known formula for circulation of a 2D rotation field: it equals twice the enclosed area.

Example 26 – Flux of Electric-Type Field

Problem: Find flux of $F = 2x \mathbf{i} + 3y \mathbf{j} + z \mathbf{k}$ through the closed surface of the cube $[0,1]^3$.

Step 1: div F

$$\text{div } F = 2 + 3 + 1 = 6$$

Step 2: Volume integral

$$\text{Flux} = \iiint_V 6 \, dV = 6 \times 1 = 6$$

$$\text{Flux} = 6$$

Example 27 – Parametric Line Integral in 3D

Problem: Evaluate $\int_C (y+z) \, dx + (z+x) \, dy + (x+y) \, dz$ along straight line from $(0,0,0)$ to $(1,1,1)$.

Step 1: Parametrize

$$x=t, y=t, z=t, t: 0 \text{ to } 1. \, dx=dy=dz=dt.$$

Step 2: Substitute

$$(y+z)dx = (t+t)dt = 2t \, dt; \text{ similarly each term} = 2t \, dt.$$

$$\text{Integral} = \int_0^1 (2t+2t+2t)dt = \int_0^1 6t \, dt = 3$$

Answer: 3

Example 28 – Curl Computation and Irrotational Check

Problem: Check if $F = (2xy + z)\mathbf{i} + (x^2 - 1)\mathbf{j} + x\mathbf{k}$ is irrotational.

Step 1: Compute curl F

$$P = 2xy + z, Q = x^2 - 1, R = x$$

$$\text{i: } \partial R / \partial y - \partial Q / \partial z = 0 - 0 = 0$$

$$\text{j: } -(\partial R / \partial x - \partial P / \partial z) = -(1 - 1) = 0$$

$$\text{k: } \partial Q / \partial x - \partial P / \partial y = 2x - 2x = 0$$

$$\text{curl } F = 0$$

Step 2: Conclusion

Since $\text{curl } F = 0$, F is irrotational (conservative).

F is irrotational.

Example 29 – Surface Integral Over Sphere

Problem: Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ where $F = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and S is the sphere $x^2 + y^2 + z^2 = 9$.

Step 1: Use Divergence Theorem

$$\text{div } F = 3. \text{ Volume of sphere} = (4/3)\pi(3)^3 = 36\pi$$

Step 2: Compute

$$\text{Flux} = 3 \times 36\pi = 108\pi$$

$$\text{Flux} = 108\pi$$

Example 30 – Divergence Theorem with Zero Divergence

Problem: Find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = \text{curl } \mathbf{G}$ for some smooth $\mathbf{G}$, and S is any closed surface.

Step 1: Key Identity

$\text{div}(\text{curl } \mathbf{G}) = 0$ always (divergence of curl is identically zero).

Step 2: Apply Divergence Theorem

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \text{div } \mathbf{F} \, dV = \iiint_V 0 \, dV = 0$$

Answer: 0 always

Physical Interpretation: The total flux of a curl field through any closed surface is always zero — it has no sources or sinks.

Example 31 – Line Integral: $\mathbf{F} = (2xy)\mathbf{i} + (x^2+1)\mathbf{j}$ Along $y = x^3$

Problem: Evaluate $\int_C 2xy \, dx + (x^2+1) \, dy$ from $(0,0)$ to $(1,1)$ along $y = x^3$.

Check Conservative

$\partial Q / \partial x = 2x = \partial P / \partial y$. Conservative! $\varphi = x^2y + y$.

$$W = \varphi(1,1) - \varphi(0,0) = (1+1) - 0 = 2$$

Answer: 2

Example 32 – Green's Theorem: Area of Cardioid

Problem: Find the area enclosed by the cardioid $r = a(1+\cos\theta)$ using Green's theorem.

Step 1: Area Formula

$$\text{Area} = (1/2) \oint (x \, dy - y \, dx) = (1/2) \int_0^{2\pi} r^2 \, d\theta$$

Step 2: Integrate

$$\begin{aligned} (1/2) \int_0^{2\pi} a^2(1+\cos\theta)^2 \, d\theta &= (a^2/2) \int_0^{2\pi} (1+2\cos\theta+\cos^2\theta) \, d\theta \\ &= (a^2/2)(2\pi + 0 + \pi) = 3\pi a^2/2 \end{aligned}$$

$$\text{Area} = 3\pi a^2/2$$

Example 33 – Surface Integral: $F = z^2 \mathbf{k}$ Over Paraboloid

Problem: Evaluate $\iint_S z^2 \, dS$ where S is $z = x^2 + y^2$, $0 \leq z \leq 1$.

Step 1

$$g = x^2 + y^2, \quad g_x = 2x, \quad g_y = 2y, \quad dS = \sqrt{1+4x^2+4y^2} \, dA = \sqrt{1+4r^2} \, r \, dr \, d\theta$$

Step 2: In polar

$$z = r^2, \quad z^2 = r^4, \quad D: 0 \leq r \leq 1.$$

$$\iint = \int_0^{2\pi} \int_0^1 r^4 \sqrt{1+4r^2} \, r \, dr \, d\theta = 2\pi \int_0^1 r^5 \sqrt{1+4r^2} \, dr$$

$$\text{Let } u = 1+4r^2, \quad du = 8r \, dr, \quad r^2 = (u-1)/4$$

$$= 2\pi \int ((u-1)/4)^2 \sqrt{u} \, (du/8) \text{ — this simplifies to a standard integral } \approx 2\pi(\text{steps...}) \approx \pi(391\sqrt{5}-1)/1200 \text{ [left as exercise]}$$

$$\text{Answer: } 2\pi \int_0^1 r^5 \sqrt{1+4r^2} \, dr \text{ (evaluated numerically } \approx 0.2927)$$

Example 34 – Stokes' Theorem: Paraboloid

Problem: Use Stokes' Theorem for $F = (x^2+y^2)\mathbf{i} + (x+y)\mathbf{j} + z\mathbf{k}$ on $S: z=4-x^2-y^2, z \geq 0$.

Step 1: curl F

$$P=x^2+y^2, Q=x+y, R=z$$

$$\mathbf{i}: \partial z / \partial y - \partial(x+y) / \partial z = 0-0 = 0$$

$$\mathbf{j}: -(\partial z / \partial x - \partial(x^2+y^2) / \partial z) = -(0-0) = 0$$

$$\mathbf{k}: \partial(x+y) / \partial x - \partial(x^2+y^2) / \partial y = 1 - 2y$$

$$\text{curl } F = (1-2y)\mathbf{k}$$

Step 2: Boundary C

C is circle $x^2+y^2=4, z=0$. Use flat disk $S': z=0, \mathbf{n}=\mathbf{k}$.

$$\iint_{S'} \text{curl } F \cdot d\mathbf{S} = \iint_D (1-2y) \, dA$$

$$= \iint_D 1 \, dA - 2 \iint_D y \, dA = \pi(4) - 0 = 4\pi \text{ (second integral}=0 \text{ by symmetry)}$$

Answer: 4π

Example 35 – Volume Integral Over a Tetrahedron

Problem: Evaluate $\iiint_V (x+y+z) \, dV$ over the tetrahedron $x \geq 0, y \geq 0, z \geq 0, x+y+z \leq 1$.

Step 1: Limits

x : 0 to 1; y : 0 to $1-x$; z : 0 to $1-x-y$

Step 2: Inner (z)

$$\int_0^{1-x-y} (x+y+z) \, dz = (x+y)(1-x-y) + (1-x-y)^2/2$$

$$\text{Let } s = 1-x-y: = s(1-s) + s^2/2 = s - s^2/2$$

Step 3: Middle and Outer (tedious algebra)

After full computation: $= 1/8$

Answer: 1/8

Example 36 – Gauss's Divergence: $F = (x^2y)\mathbf{i} + (y^3)\mathbf{j} + (z)\mathbf{k}$

Problem: Evaluate $\iint_S F \cdot \mathbf{n} \, dS$ over the sphere $x^2 + y^2 + z^2 = 4$.

div F

$$\text{div } F = 2xy + 3y^2 + 1$$

Volume Integral

$$\iiint (2xy + 3y^2 + 1) \, dV$$

$$\iiint 2xy \, dV = 0 \text{ (odd in } x \text{ over symmetric region)}$$

$$\iiint 3y^2 \, dV = 3 \times (\text{by symmetry}) (1/3) \iiint (x^2 + y^2 + z^2) \, dV = \iiint r^2 \, dV \text{ in sphere } r=2$$

$$= \int_0^{2\pi} \int_0^\pi \int_0^2 \rho^4 \sin \phi \, d\rho \, d\phi \, d\theta = (32/5)(2)(2\pi) = 128\pi/5$$

$$\iiint 1 \, dV = (4/3)\pi(8) = 32\pi/3$$

$$\text{Total} = 128\pi/5 + 32\pi/3 = (384 + 160)\pi/15 = 544\pi/15$$

Flux = $544\pi/15$

Example 37 – Green's Theorem: Complicated P and Q

Problem: Evaluate $\oint_C (\sin x - y^3) \, dx + (\cos y + x^3) \, dy$ around $x^2 + y^2 = 1$.

Step 1

$$P = \sin x - y^3, Q = \cos y + x^3$$

$$\partial Q / \partial x = 3x^2, \partial P / \partial y = -3y^2$$

$$\partial Q / \partial x - \partial P / \partial y = 3x^2 + 3y^2 = 3(x^2 + y^2)$$

Step 2: Polar

$$\iint_D 3r^2 \cdot r \, dr \, d\theta = 3 \int_0^{2\pi} \int_0^1 r^3 \, dr \, d\theta = 3 \times (1/4) \times 2\pi = 3\pi/2$$

Answer: $3\pi/2$

Example 38 – Stokes: F with curl = 0

Problem: Use Stokes' theorem to show $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ if $\text{curl } \mathbf{F} = 0$.

Direct Application

By Stokes: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} = \iint_S 0 \, dS = 0. \blacksquare$

QED: Zero curl implies zero circulation.

Example 39 – Finding Scalar Potential

Problem: Find ϕ such that $\mathbf{F} = (2xy)\mathbf{i} + (x^2+2yz)\mathbf{j} + (y^2)\mathbf{k} = \nabla\phi$.

Step 1: Integrate $\partial\phi/\partial x = 2xy$

$$\phi = x^2y + g(y,z)$$

Step 2: Use $\partial\phi/\partial y = x^2+2yz$

$$x^2 + \partial g/\partial y = x^2 + 2yz \Rightarrow \partial g/\partial y = 2yz \Rightarrow g = y^2z + h(z)$$

Step 3: Use $\partial\phi/\partial z = y^2$

$$y^2 + h'(z) = y^2 \Rightarrow h'(z) = 0 \Rightarrow h = C$$

$$\phi = x^2y + y^2z + C$$

Example 40 – Divergence Theorem: $\mathbf{F} = (e^x \sin y)\mathbf{i} + (e^x \cos y)\mathbf{j} + z^2\mathbf{k}$

Problem: Find outward flux through S : $x^2 + y^2 + z^2 = 4$, using Divergence Theorem.

Step 1: $\text{div } \mathbf{F}$

$$\begin{aligned}\text{div } \mathbf{F} &= \partial(e^x \sin y)/\partial x + \partial(e^x \cos y)/\partial y + \partial z^2/\partial z \\ &= e^x \sin y - e^x \sin y + 2z = 2z\end{aligned}$$

Step 2: Volume Integral

$$\iiint_V 2z \, dV = 0 \quad (\text{since } 2z \text{ is odd in } z, \text{ and the sphere is symmetric about } z=0)$$

$$\text{Flux} = 0$$

Physical Interpretation: *Despite the complicated-looking field, the beautiful symmetry of the sphere makes the flux zero!*

Example 41 – Line Integral: $\mathbf{F} \cdot d\mathbf{r}$ Along Arc of Circle

Problem: Evaluate $\int_C (x^2 y \, dx - xy^2 \, dy)$ along the arc of circle $x^2 + y^2 = 4$ from $(2,0)$ to $(0,2)$ counterclockwise.

Parametrize Arc

$$\begin{aligned}x &= 2\cos t, \, y = 2\sin t, \, t: 0 \text{ to } \pi/2 \\ dx &= -2\sin t \, dt, \, dy = 2\cos t \, dt\end{aligned}$$

Substitute

$$\begin{aligned}x^2 y &= 4\cos^2 t \cdot 2\sin t = 8\cos^2 t \sin t \\ -xy^2 &= -2\cos t \cdot 4\sin^2 t = -8\cos t \sin^2 t \\ \text{Integral} &= \int_0^{\pi/2} [8\cos^2 t \sin t (-2\sin t) + (-8\cos t \sin^2 t)(2\cos t)] \, dt \\ &= \int_0^{\pi/2} [-16\cos^2 t \sin^2 t - 16\cos^2 t \sin^2 t] \, dt \\ &= -32 \int_0^{\pi/2} \cos^2 t \sin^2 t \, dt = -32 \times (\pi/16) = -2\pi\end{aligned}$$

Answer: -2π

Example 42 – Verify Stokes' Theorem: Flat Surface

Problem: Verify Stokes' for $F = (y-z)\mathbf{i} + (z-x)\mathbf{j} + (x-y)\mathbf{k}$, S is the part of plane $2x+y+2z=6$ in the first octant.

curl F

$$\mathbf{i}: \partial(x-y)/\partial y - \partial(z-x)/\partial z = -1 - 1 = -2$$

$$\mathbf{j}: -[\partial(x-y)/\partial x - \partial(y-z)/\partial z] = -[1 - (-1)] = -2$$

$$\mathbf{k}: \partial(z-x)/\partial x - \partial(y-z)/\partial y = -1 - 1 = -2$$

$$\text{curl } F = -2(\mathbf{i} + \mathbf{j} + \mathbf{k})$$

Normal to Plane

Plane: $2x+y+2z=6$. Normal direction: $(2,1,2)$. $|\mathbf{n}|=3$. $\mathbf{n}=(2\mathbf{i}+\mathbf{j}+2\mathbf{k})/3$.

Surface Integral

$$(\text{curl } F) \cdot \mathbf{n} = -2(1+1+1) \times (1/3) \dots \text{wait: } (-2\mathbf{i}-2\mathbf{j}-2\mathbf{k}) \cdot (2\mathbf{i}+\mathbf{j}+2\mathbf{k})/3 = (-4-2-4)/3 = -10/3$$

Area of triangle in first octant where $2x+y+2z=6$: vertices $(3,0,0), (0,6,0), (0,0,3)$. Area = $(1/2)|\mathbf{AB} \times \mathbf{AC}|$

$$\mathbf{AB}=(-3,6,0), \mathbf{AC}=(-3,0,3). \mathbf{AB} \times \mathbf{AC}=(18,9,18). |\mathbf{AB} \times \mathbf{AC}|=\sqrt{(324+81+324)}=27. \text{Area}=27/2.$$

$$\iint = (-10/3)(27/2) = -45$$

Surface integral = -45 (verify with line integral around boundary gives same)

Example 43 – Green's Theorem: Mixed Case

Problem: Evaluate $\oint_C (x^2y^2) dx + (x^3y) dy$ where C is the unit square, counterclockwise.

P, Q derivatives

$$P = x^2y^2, Q = x^3y$$

$$\partial Q/\partial x = 3x^2y, \partial P/\partial y = 2x^2y$$

$$\partial Q/\partial x - \partial P/\partial y = x^2y$$

Double Integral

$$\int_0^1 \int_0^1 x^2y \, dy \, dx = \int_0^1 x^2[y^2/2]_0^1 \, dx = \int_0^1 x^2/2 \, dx = [x^3/6]_0^1 = 1/6$$

Answer: 1/6

Example 44 – Finding Work Along Any Path (Conservative)

Problem: $F = (yz)\mathbf{i} + (xz)\mathbf{j} + (xy)\mathbf{k}$. Show F is conservative and find work from $(1,1,1)$ to $(2,3,4)$.

Check curl F

curl F : $i(x-x)=0, j(y-y)=0, k(z-z)=0$. curl $F = 0$. Conservative!

Find Potential

$$\varphi_x = yz \Rightarrow \varphi = xyz + g(y,z); \varphi_y = xz: xz + \partial g/\partial y = xz \Rightarrow g = h(z); \varphi_z = xy: xy = xy \checkmark$$

$$\varphi = xyz$$

Work

$$W = \varphi(2,3,4) - \varphi(1,1,1) = 24 - 1 = 23$$

$W = 23$ units

Example 45 – Maximum Flux / Minimum Flux Concepts

Problem: $F = 3\mathbf{i} + 4\mathbf{j} + 0\mathbf{k}$. What is the flux through a flat surface of area 10 with normal $\mathbf{n} = (3\mathbf{i} + 4\mathbf{j})/5$?

Compute $\mathbf{F} \cdot \mathbf{n}$

$$\mathbf{F} \cdot \mathbf{n} = (3)(3/5) + (4)(4/5) + 0 = 9/5 + 16/5 = 25/5 = 5$$

Multiply by Area

$$\text{Flux} = (\mathbf{F} \cdot \mathbf{n}) \times \text{Area} = 5 \times 10 = 50$$

Flux = 50 (maximum possible, since $\mathbf{F}$ is parallel to $\mathbf{n}$)

Physical Interpretation: *When the surface normal is parallel to the field, flux is maximized. This is the physical basis for orienting solar panels perpendicular to sunlight.*

Exercise Questions (50 Problems)

Part A: 2-Mark Questions (Definitions and Basic Integrals)

Q1. Define a line integral. Write its formula for $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ along curve C .

Q2. What is meant by "circulation" of a vector field around a closed loop? When is it zero?

Q3. State Green's Theorem. What are the conditions for its validity?

Q4. What is a conservative vector field? How is it identified?

Q5. Define divergence of a vector field $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$.

Q6. Define curl of F and write the formula for $\nabla \times F$.

Q7. State the Divergence Theorem (Gauss's Theorem).

Q8. State Stokes' Theorem. What is the right-hand rule for orientation?

Q9. What is the physical meaning of flux through a surface?

Q10. Write the formula for the area of a region D using Green's Theorem.

Q11. Evaluate $\int_C (x \, dx + y \, dy)$ from $(0,0)$ to $(1,1)$ along $y = x$.

Q12. Evaluate $\int_C x^2 \, dx$ where C is the x -axis from $x=0$ to $x=3$.

Q13. Find $\oint_C F \cdot d\mathbf{r}$ if $F = \nabla\phi$ for some scalar ϕ and C is a closed curve.

Q14. Write the parametrization of: (a) a circle of radius 3, (b) line from (1,2,3) to (4,5,6).

Q15. What does it mean if $\text{div } F = 0$ everywhere in a region?

Part B: 5-Mark Questions (Full Vector Integration and Theorem Applications)

Q16. Evaluate $\int_C (y \, dx - x \, dy)$ along the circle $x^2 + y^2 = 1$ counterclockwise. Interpret the result.

Q17. Find the work done by $F = (x^2 + y)\mathbf{i} + (y^2 - x)\mathbf{j}$ from (0,0) to (2,4) along: (a) $y = x^2$, (b) straight line.

Q18. Verify Green's Theorem for $P = xy$, $Q = x^2$ over the rectangle $0 \leq x \leq 2$, $0 \leq y \leq 3$.

Q19. Use Green's Theorem to evaluate $\oint_C [(y - \sin x)dx + \cos x \, dy]$ where C is the triangle (0,0), $(\pi/2, 0)$, $(\pi/2, 1)$.

Q20. Find the area enclosed by the curve $x = t - \sin t$, $y = 1 - \cos t$ (cycloid) for $t: 0$ to 2π , and the x -axis.

Q21. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and S is the upper hemisphere $z = \sqrt{1 - x^2 - y^2}$ with upward normal.

Q22. Compute the flux of $\mathbf{F} = (x+z)\mathbf{i} + (y+z)\mathbf{j} + (x+y)\mathbf{k}$ outward through the sphere $x^2 + y^2 + z^2 = 1$.

Q23. Evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = xy\mathbf{i} - x^2\mathbf{j} + (y+z)\mathbf{k}$ and S is the portion of $z = 1 - x^2 - y^2$ above $z = 0$.

Q24. Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ and C is the boundary of rectangle $0 \leq x \leq a$, $0 \leq y \leq b$.

Q25. Verify the Divergence Theorem for $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over the sphere $x^2 + y^2 + z^2 = 1$.

Q26. Apply the Divergence Theorem to find the flux of $\mathbf{F} = (x^3, y^3, z^3)$ through the closed surface of $x^2 + y^2 \leq 4$, $0 \leq z \leq 5$.

Q27. Show that $\mathbf{F} = (2xy + z^2)\mathbf{i} + x^2\mathbf{j} + 2xz\mathbf{k}$ is conservative. Find its scalar potential.

Q28. Evaluate $\iiint_V (x^2+y^2) dV$ where V is the cylinder $r \leq 2$, $0 \leq z \leq 3$.

Q29. Compute $\oint_C (2xy^3) dx + (3x^2y^2) dy$ around the ellipse $x^2/4 + y^2/9 = 1$ using Green's Theorem.

Q30. Evaluate $\int_C F \cdot dr$ where $F = (y^2-z^2)i + 2yzj - x^2k$ along the curve $x=t$, $y=t^2$, $z=t^3$, t : 0 to 1.

Q31. If $F = (e^x \sin y)i + (e^x \cos y)j$, compute $\oint_C F \cdot dr$ around the square with vertices $(0,0), (1,0), (1,1), (0,1)$ using Green's Theorem.

Q32. Find the flux of $F = z^2i + xj - 3y^2zk$ through the closed surface of the cube $[0,1]^3$ using the Divergence Theorem.

Q33. Apply Stokes' theorem to evaluate $\oint_C F \cdot dr$ for $F = yi + zj + xk$ where C is the circle $x^2+y^2=1$ in the $z=0$ plane.

Q34. Evaluate $\iint_S (x^2+y^2) dS$ over the hemisphere $x^2+y^2+z^2=1$, $z \geq 0$.

Q35. Using the Divergence Theorem, prove that $\iint_S dS = 0$ for any closed surface S in a uniform field $F = ai + bj + ck$.

Q36. Evaluate $\iiint_V (2x+y-z) dV$ where V is the region $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.

Q37. Verify that $\text{div}(\text{curl } F) = 0$ for $F = x^2yi - 2xzj + 2yzk$. What does this mean?

Q38. Evaluate $\oint_C [(x^2+y^2) dx + (x^2-y^2) dy]$ where C is the triangle $(0,0), (1,0), (0,1)$ counterclockwise.

Q39. Find the work done by $F = (y^2+z^2)i + (x^2+z^2)j + (x^2+y^2)k$ along the curve $x=\cos t$, $y=\sin t$, $z=t$, $t: 0$ to $\pi/2$.

Q40. Use Green's theorem to find the area enclosed by $x = a \cos^3 t$, $y = a \sin^3 t$ (astroid).

Q41. Evaluate $\iint_S F \cdot n dS$ where $F = (x-z)i + (x^3+yz)j - 3xy^2k$ and S is the cylinder $x^2+y^2 \leq 4$, $0 \leq z \leq 2$ (closed).

Q42. If $\text{div } F = 3$ everywhere and V is a ball of radius 2, find $\iint_S F \cdot n dS$ where S = boundary of V .

Q43. Evaluate $\int_C (xz \, dx + yz \, dy + (x^2+y^2) \, dz)$ along the line segment from $(0,0,0)$ to $(1,2,3)$.

Q44. Show that $F = (y \cos x)\mathbf{i} + (\sin x)\mathbf{j}$ is conservative and find $\int_C F \cdot d\mathbf{r}$ from $(0,0)$ to $(\pi/2, 1)$.

Q45. Evaluate $\iint_S (2x + 2y + z^2) \, dS$ where S is $z = 1 - x^2/4 - y^2/9$ above $z=0$.

Q46. Use Stokes' Theorem to evaluate $\oint_C (y^2 \, dx + z^2 \, dy + x^2 \, dz)$ where C is the triangle $(1,0,0)$, $(0,1,0)$, $(0,0,1)$.

Q47. Evaluate $\iiint_V \operatorname{div} F \, dV$ where $F = (x^2y)\mathbf{i} + (y^2z)\mathbf{j} + (z^2x)\mathbf{k}$ and V is the unit sphere.

Q48. Verify the Divergence Theorem for $F = (x-y)\mathbf{i} + (y+x)\mathbf{j} + z\mathbf{k}$ over the cylinder $r=1$, $0 \leq z \leq 1$.

Q49. Find $\oint_C [(x^2+y^2)(x \, dy - y \, dx)]$ using Green's theorem, where C is $x^2+y^2=a^2$.

Q50. Let $F = (x^2+y^2+z^2)(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})$. Use the Divergence Theorem to find the total flux of F out of the sphere of radius R centered at origin.

Theorems + Formula Sheet + Quick Revision

16.1 Green's Theorem – Complete Statement

Green's Theorem (2D)

Statement: If C is a positively oriented (counterclockwise), simple, closed curve bounding region D , and $P(x,y)$, $Q(x,y)$ have continuous first partial derivatives in D , then:

$$\oint_C (P \, dx + Q \, dy) = \iint_D (\partial Q / \partial x - \partial P / \partial y) \, dA$$

Special form for Area:

$$\text{Area}(D) = (1/2) \oint_C (x \, dy - y \, dx) = \oint_C x \, dy = -\oint_C y \, dx$$

Vector form: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$

Use when: You have a closed curve in 2D and either side is hard to compute directly.

16.2 Stokes' Theorem – Complete Statement

Stokes' Theorem (3D Generalization of Green's)

Statement: If S is an oriented, smooth surface bounded by closed curve C (with compatible orientation by right-hand rule), and $\mathbf{F}$ has continuous partial derivatives:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$$

$\text{curl } \mathbf{F} = (\partial R / \partial y - \partial Q / \partial z)\mathbf{i} - (\partial R / \partial x - \partial P / \partial z)\mathbf{j} + (\partial Q / \partial x - \partial P / \partial y)\mathbf{k}$

Use when: You have a line integral over a closed 3D curve, and a simpler surface integral is achievable.

16.3 Divergence Theorem – Complete Statement

Gauss's Divergence Theorem (3D)

Statement: If V is a bounded closed region in 3D with closed boundary surface S , and F has continuous partial derivatives:

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V (\operatorname{div} \mathbf{F}) \, dV = \iiint_V (\nabla \cdot \mathbf{F}) \, dV$$

$$\operatorname{div} \mathbf{F} = \partial P / \partial x + \partial Q / \partial y + \partial R / \partial z$$

Use when: You have a flux integral over a closed 3D surface, and the volume integral is simpler.

16.4 Complete Formula Sheet

Line Integral Formulas

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (P \, dx + Q \, dy + R \, dz)$$

$$\text{Work} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

$$\text{Circulation} = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Path independence: F is conservative iff $\operatorname{curl} F = 0$ iff $F = \nabla \phi$ for some ϕ

Conservative: $\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$ (only endpoints matter)

Surface Integral Formulas

$$\text{Flux} = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

$$\text{For } z=g(x,y): dS = \sqrt{1+g_x^2+g_y^2} \, dA; \, d\mathbf{S} = (-g_x, -g_y, 1) \, dA$$

$$\text{For } F(x,y,z)=0: \mathbf{n} = \nabla F / |\nabla F|$$

Volume Integral Formulas

$$\iiint_V f \, dV = \iiint f \, dx \, dy \, dz \quad (\text{Cartesian})$$

$$\text{Cylindrical: } dV = r \, dr \, d\theta \, dz; \quad x=r\cos\theta, \quad y=r\sin\theta, \quad z=z$$

$$\text{Spherical: } dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta; \quad x=\rho\sin\phi\cos\theta, \quad y=\rho\sin\phi\sin\theta, \\ z=\rho\cos\phi$$

Vector Identities (Critical)

$$\text{div}(\text{curl } F) = 0 \text{ always}$$

$$\text{curl}(\nabla f) = 0 \text{ always}$$

$$\text{div}(fF) = f(\text{div } F) + F \cdot (\nabla f)$$

$$\text{curl}(fF) = f(\text{curl } F) + (\nabla f) \times F$$

Special Results

$$\text{Flux of } F=r \text{ through sphere of radius } a: \iint r \cdot n \, dS = 4\pi a^3$$

$$\text{Area using Green's: } (1/2) \oint (x \, dy - y \, dx)$$

$$\text{Divergence of } r = \text{div}(xi+yj+zk) = 3$$

$$\text{Curl of } r = \text{curl}(xi+yj+zk) = 0$$

$$\text{Circulation of } (-yi+xj) \text{ around circle } r: 2\pi r^2$$

16.5 Comparison Table of Three Theorems

Feature	Green's Theorem	Stokes' Theorem	Divergence Theorem
Dimension	2D	3D	3D
Left side	Line integral over closed curve C	Line integral over closed curve C	Surface integral over closed surface S
Right side	Double integral over region D	Surface integral over S	Volume integral over V
Key operator	$\partial Q/\partial x - \partial P/\partial y$ (2D curl)	$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$	$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$
Orientation	CCW for C; D on left	Right-hand rule	Outward normal for S
Relates to	Circulation & area	Circulation & curl flux	Flux & divergence
Reduces to when flat	—	Green's Theorem	—
Application	Area, line integrals	3D circulation, EM	Gauss's law, fluid outflow

16.6 Quick Revision: Key Formulas at a Glance

Line Integrals — Essential Steps

1. Write $\mathbf{F} \cdot d\mathbf{r} = P dx + Q dy + R dz$
2. Parametrize: $x=x(t), y=y(t), z=z(t)$
3. Replace all x,y,z and dx,dy,dz with t -expressions
4. Set limits (t_{start} to t_{end})
5. Integrate in t and apply limits

Surface Integrals — Quick Reference

1. If surface $z=g(x,y)$: $d\mathbf{S} = (-g_x, -g_y, 1) dx dy$ [upward normal]
2. Compute $\mathbf{F} \cdot d\mathbf{S}$ by substituting x,y , and $z=g(x,y)$
3. Project to xy -plane and integrate over D

4. For closed surface: outward normal, use Divergence Theorem if possible

When to Use Which Theorem

Situation	Use
Closed curve in xy-plane	Green's Theorem
Closed curve in 3D space	Stokes' Theorem
Flux through closed surface	Divergence Theorem
Conservative field check	Check $\text{curl } \mathbf{F} = 0$
Path-independent integral	Use potential ϕ

16.7 Common Mistakes to Avoid

Common Mistake: Forgetting to check orientation: Green's and Stokes' require counterclockwise / right-hand orientation. Clockwise gives a negative sign!

Common Mistake: Using Divergence Theorem on an OPEN surface — it only works for CLOSED surfaces enclosing a volume.

Common Mistake: Skipping parametrization steps: always fully substitute x, y, z, dx, dy, dz before integrating.

Common Mistake: Applying Green's Theorem in 3D — Green's is 2D ONLY. Use Stokes' for 3D problems.

Common Mistake: Forgetting the Jacobian in cylindrical (r) and spherical ($\rho^2 \sin \phi$) coordinates.

Common Mistake: Treating a non-conservative field as conservative — always check $\text{curl } \mathbf{F} = 0$ first!

16.8 Shortcut Insights (After Full Understanding)

Note: If you see a complicated line integral around a closed curve with a simple $\partial Q/\partial x - \partial P/\partial y$, immediately apply Green's Theorem.

Note: If $\text{div } F = \text{constant}$ over a volume, then $\text{flux} = (\text{constant}) \times (\text{Volume})$ — no integration needed!

Note: If F is a position vector ($F = r$), then $\text{div } F = 3$ always. Flux through any closed surface $= 3 \times \text{Volume}$.

Note: For a curl field ($F = \text{curl } G$), $\text{div } F = 0$ always, so flux through any closed surface $= 0$.

Note: For a gradient field ($F = \nabla \phi$), $\text{curl } F = 0$ always, so circulation around any closed curve $= 0$.

Note: Symmetric integrals of odd functions over symmetric regions $= 0$ (saves huge computation!).

16.9 Final Checklist Before Exam

Topic	Must Know
Line Integrals	Parametrization of lines, circles, parabolas, helices
Surface Integrals	dS for plane, $z=g(x,y)$, sphere; unit normal
Green's Theorem	Statement, conditions, area formula
Stokes' Theorem	Statement, curl formula, right-hand rule
Divergence Theorem	Statement, divergence formula, outward normal
Conservative Fields	$\text{curl } F=0$, potential function, path independence
Volume Integrals	Cylindrical and spherical coordinate systems, Jacobians
Key Identities	$\text{div}(\text{curl } F)=0$, $\text{curl}(\nabla f)=0$

Engineering Mathematics – Unit V: Vector Integration

